

SOFTWARE VERIFICATION AND VALIDATION
SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

TOPIC:

" COMPLAINT MANAGMENT SYSTEM (TICKET-BASED WORKFLOW) "



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COMPLAINT MANAGEMENT SYSTEM (TICKET-BASED WORKFLOW)

1. Introduction

1.1 Purpose

The purpose of this system is to provide a structured and reliable way to manage **complaints using a ticket-based** approach. In many organizations, complaints are handled manually which leads to delays, confusion, and sometimes loss of information.

This system allows users to create complaints in the form of tickets, which are handled by support agents and managed by admin. Each complaint follows a proper lifecycle, ensuring that it is handled efficiently.

1.2 Scope

The Complaint Management System manages complaints from **creation to closure**. It supports multiple roles including **user, support agent, and admin**.

The system includes:

- Ticket creation
- Ticket assignment
- Complaint handling
- Status tracking
- Resolution and closure

The system does not include advanced features like payments, notifications, or external integrations.

1.3 Definitions

- **User:** A person who submits a complaint
- **Support Agent:** A person who resolves complaints
- **Admin:** A person who manages and assigns tickets
- **Ticket:** A complaint record

- **Status:** Current state of a ticket
- **Resolution:** Solution provided for a complaint

1.4 Document Overview

This document describes all **system requirements** including functional and non-functional requirements, system behavior, and validation conditions.

1.5 Objectives

The main objectives of the system are:

- To provide an easy and structured way for users to submit complaints
- To ensure all complaints are tracked using tickets
- To allow support agents to resolve complaints efficiently
- To provide clear visibility of complaint status
- To reduce manual effort and improve organization
- To ensure no complaint is lost or ignored

2. Overall Description

2.1 System Perspective

This is a standalone system used to manage complaints internally.

Flow:



2.2 System Functions

- Create ticket
- Assign ticket
- Update ticket status
- Resolve complaint
- Close ticket
- Reopen ticket

2.3 User Classes and Characteristics

User

- Creates tickets
- Views ticket status

Support Agent

- Handles assigned tickets
- Resolves complaints
- Updates status

Admin

- Assigns tickets
- Monitors system
- Controls workflow

2.4 Operating Environment

- Web-based system
- Runs on modern browsers
- Accessible via computer/laptop
- Requires internet

2.5 Assumptions

- Users have internet access
- Support agents are trained
- Admin has full authority
- Limited number of users

2.6 Constraints

- Only basic workflow implemented
- No mobile application
- No external system integration

3. System Workflow

3.1 Ticket Lifecycle

1. Open → Ticket created
2. Assigned → Assigned by admin
3. In Progress → Agent working
4. Resolved → Solution provided
5. Closed → Ticket closed
6. Reopened → If needed

3.2 Workflow Rules

- Ticket must follow defined sequence
- Ticket cannot be closed without resolution
- Reopened ticket returns to active state

4. Functional Requirements

Functional Requirement ID	Requirement Name	Description
FR1	Create Ticket	User can create a ticket with a complaint description. Default ticket status is Open .
FR2	Assign Ticket	Admin assigns the ticket to a support agent. Ticket status changes to Assigned .
FR3	Start Processing Ticket	Support agent starts working on the ticket. Ticket status changes to In Progress .
FR4	Resolve Ticket	Support agent adds the resolution details. Ticket status changes to Resolved .
FR5	Close Ticket	After resolution, the ticket is officially closed. Ticket status becomes Closed .
FR6	Reopen Ticket	A closed ticket can be reopened if the issue still exists. Ticket status changes back to an active state.

5. Non-Functional Requirements

Non-Functional Requirement	Description
Usability	The system should be simple, easy to use, and user-friendly for all users.
Performance	The system should respond within 2–3 seconds and handle multiple tickets efficiently.
Security	Only authorized users should perform actions. Role-based access control must be implemented.
Reliability	The system should not lose data, and all tickets must be stored safely and securely.
Maintainability	The system should be easy to update, maintain, and manage in the future.

6. Data Requirements

Each ticket includes:

- Ticket ID (unique)
- Complaint description
- Status
- Assigned agent
- Resolution notes

7. System Invariants

1. Every ticket must have a unique ID
2. Closed tickets cannot be reassigned
3. Resolved tickets must contain resolution notes

8. Requirement Defect Table

Defect ID	Requirement	Issue Type	Description
D1	Ticket details	Ambiguity	Required fields not clearly defined
D2	Reopen ticket	Incomplete	Conditions for reopening unclear
D3	Role permissions	Missing	Access control not fully specified

9. Validation Checklist

Requirement	Verified
Create ticket	Yes
Assign ticket	Yes
Resolve ticket	Yes
Close ticket	Yes
Reopen ticket	Yes

10. Conclusion

The Complaint Management System provides a structured and efficient way to manage complaints using a ticket-based workflow. It ensures proper tracking, handling, and resolution of complaints.

The system reduces manual work, improves organization, and provides better communication between users, support agents, and admin.